EX NO:2 Develop and compare CLI, GUI, and VOI Interfaces (VUI) using python

i) CLI (Command Line Interface)

CLI implementation where users can add, view, and remove tasks using the terminal.

```
tasks = []
def add_task(task):
  tasks.append(task)
  print(f"Task '{task}' added.")
def view_tasks():
  if tasks:
    print("Your tasks:")
    for idx, task in enumerate(tasks, 1):
      print(f"{idx}. {task}")
  else:
    print("No tasks to show.")
def remove_task(task_number):
  if 0 < task_number <= len(tasks):
    removed_task = tasks.pop(task_number - 1)
    print(f"Task '{removed_task}' removed.")
  else:
    print("Invalid task number.")
def main():
  while True:
    print("\nOptions: 1.Add Task 2.View Tasks 3.Remove Task 4.Exit"
    choice = input("Enter your choice: ")
     if choice == '1':
      task = input("Enter task: ")
      add_task(task)
```

```
        elif choice == '2':
            view_tasks()

        elif choice == '3':
            task_number = int(input("Enter task number to remove: "))
            remove_task(task_number)

        elif choice == '4':
            print("Exiting...")
            break

        else:
            print("Invalid choice. Please try again.")

if __name__ == "__main__":
    main()
```

OUTPUT:

```
Options: 1.Add Task 2.View Tasks 3.Remove Task 4.Exit
Enter your choice: 1
Enter task: prepare for tommorow's exam
Task 'prepare for tommorow's exam' added.

Options: 1.Add Task 2.View Tasks 3.Remove Task 4.Exit
Enter your choice: 1
Enter task: revise 2 units in 2 hours
Task 'revise 2 units in 2 hours' added.

Options: 1.Add Task 2.View Tasks 3.Remove Task 4.Exit
Enter your choice: take 5 mins break
Invalid choice. Please try again.

Options: 1.Add Task 2.View Tasks 3.Remove Task 4.Exit
Enter your choice: 1
Enter task: take 5 mins break
Task 'take 5 mins break ' added.

Options: 1.Add Task 2.View Tasks 3.Remove Task 4.Exit
Enter your choice: 2
Your tasks:
1. prepare for tommorow's exam
2. revise 2 units in 2 hours
3. take 5 mins break

Options: 1.Add Task 2.View Tasks 3.Remove Task 4.Exit
Enter your choice: 3
Enter task number to remove: 3
Task 'take 5 mins break ' removed.

Options: 1.Add Task 2.View Tasks 3.Remove Task 4.Exit
Enter your choice: 2
Your tasks:
1. prepare for tommorow's exam
2. revise 2 units in 2 hours

Options: 1.Add Task 2.View Tasks 3.Remove Task 4.Exit
Enter your choice:
```

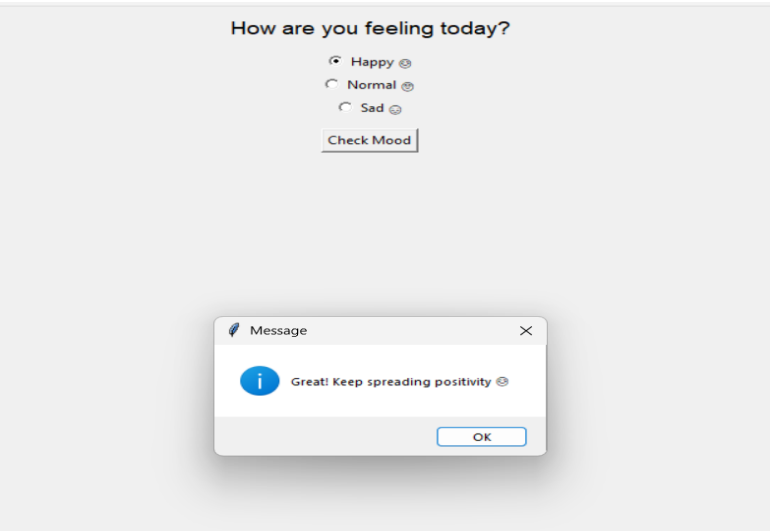
ii) GUI (Graphical User Interface)

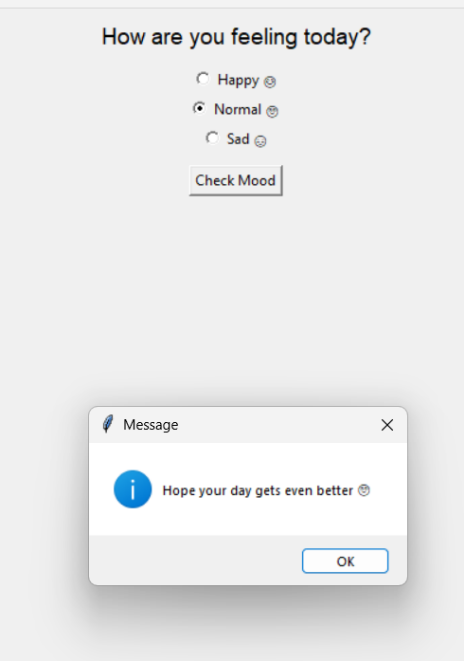
```
import tkinter as tk
```

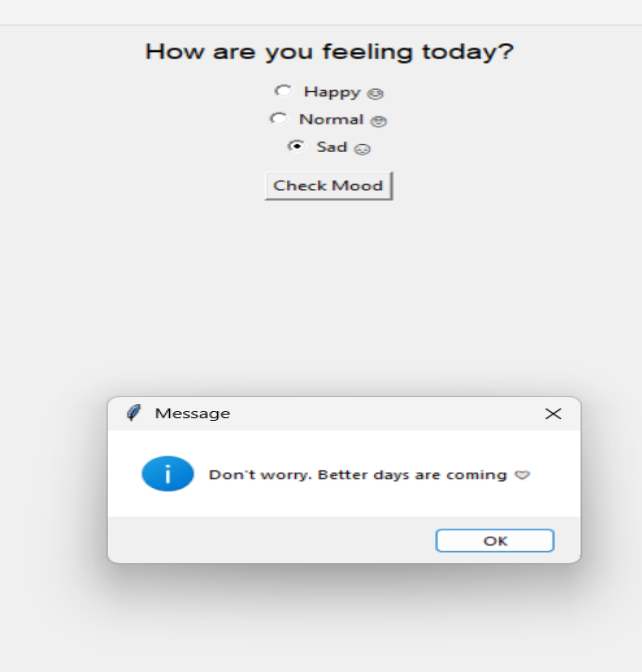
```
from tkinter import messagebox
def check_mood():
   mood = mood_var.get()
    if mood == "Happy":
      messagebox.showinfo("Message", "Great! Keep spreading positivity 😊 ")
   elif mood == "Normal":
      messagebox.showinfo("Message", "Hope your day gets even better 🙂 ")
   elif mood == "Sad":
      messagebox.showinfo("Message", "Don't worry. Better days are coming 💛 ")

   else:
      messagebox.showwarning("Warning", "Please select your mood")
app = tk.Tk()
app.title("Mood Checker")
label = tk.Label(app, text="How are you feeling today?", font=("Arial", 14))
label.pack(pady=10)
mood_var = tk.StringVar()
happy = tk.Radiobutton(app, text="Happy 😊 ", variable=mood_var, value="Happy")
happy.pack()
normal = tk.Radiobutton(app, text="Normal 🙂 ", variable=mood_var, value="Normal")
normal.pack()
sad = tk.Radiobutton(app, text="Sad 😔 ", variable=mood_var, value="Sad")
sad.pack()
check_button = tk.Button(app, text="Check Mood", command=check_mood)
check_button.pack(pady=10)
app.mainloop()
```

Output:







iii) VUI (Voice User Interface)

```
import speech_recognition as sr

import pyttsx3

tasks = []

recognizer = sr.Recognizer()

engine = pyttsx3.init()
```

```
def add_task(task):
    tasks.append(task)
    engine.say(f"Task {task} added")
    engine.runAndWait()
def view_tasks():
    if tasks:
        engine.say("Your tasks are")
        for task in tasks:
            engine.say(task)
    else:
        engine.say("No tasks to show")
    engine.runAndWait()
def remove_task(task_number):
    if 0 < task_number <= len(tasks):
        removed_task = tasks.pop(task_number - 1)
        engine.say(f"Task {removed_task} removed")
    else:
        engine.say("Invalid task number")
    engine.runAndWait()
def recognize_speech():
    with sr.Microphone() as source:
        print("Listening...")
        audio = recognizer.listen(source)
    try:
        command = recognizer.recognize_google(audio)
        print("You said:", command)
        return command.lower()
    except sr.UnknownValueError:
        engine.say("Sorry, I did not understand that")
```

```
    engine.runAndWait()
    return None
def main():
  while True:
    engine.say("Options are add task, view tasks, remove task, or exit")
    engine.runAndWait()
    command = recognize_speech()
    if not command:
      continue
    if "add task" in command:
      engine.say("What is the task?")
      engine.runAndWait()
      task = recognize_speech()
      if task:
        add_task(task)
    elif "view tasks" in command:
      view_tasks()
    elif "remove task" in command:
      engine.say("Which task number to remove?")
      engine.runAndWait()
      task_number = recognize_speech()
      if task_number:
        try:
          remove_task(int(task_number))
        except ValueError:
          engine.say("Please say a number")
          engine.runAndWait()
    elif "exit" in command:
      engine.say("Exiting")
```

```
        engine.runAndWait()
        break
    else:
        engine.say("Invalid option, please try again")
        engine.runAndWait()

if __name__ == "__main__":
  main()
```

Output:

```
(Voice) Options: add task, view tasks, remove task, or exit
(User) add task
(Voice) What is the task?
(User) Finish report
(Voice) Task Finish report added
```

```
(Voice) Options: add task, view tasks, remove task, or exit
(User) view tasks
(Voice) Your tasks are: Finish report
```

```
(Voice) Options: add task, view tasks, remove task, or exit
(User) remove task
(Voice) Which task number to remove?
(User) 1
(Voice) Task Finish report removed
```